WORKED SOLUTION

STEP II 2004, Question 14

Probability and Statistics • Inclusion-Exclusion Principle

STEP II 2004, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q14

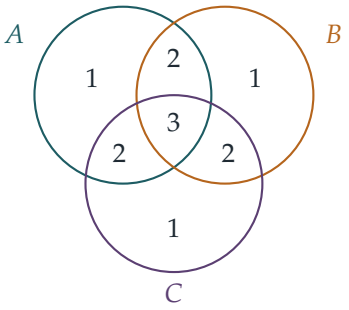
The key idea

Inclusion–exclusion is a counting correction: adding $\mathrm{P}(A)+\mathrm{P}(B)+\mathrm{P}(C)$ counts overlaps too often, and the pairwise and triple terms put the count right. For the puddings it is much easier to work with the *complement*: let E_i be the event that pudding i is empty. The probability that *some* pudding is empty is a union, which is exactly what inclusion–exclusion handles, and each intersection of the E_i has a very simple probability.

Solution

The inclusion–exclusion formula

Split $A \cup B \cup C$ into the seven disjoint regions of the Venn diagram. The probability of any event made of whole regions is the sum of the probabilities of its regions, so it is enough to check that each region is counted exactly once on the right-hand side.



The numbers in the diagram show how many of A, B, C each region lies in. Take a region lying in exactly k of the three events, where $k = 1$, 2 or 3.

- It appears in k of the terms $\mathrm{P}(A), \mathrm{P}(B), \mathrm{P}(C)$.

- It appears in $\binom{k}{2}$ of the pairwise intersections, because it lies in $A \cap B$, say, exactly when it lies in both A and B.
- It appears in the triple intersection only if $k = 3$, that is, $\binom{k}{3}$ times.

So the net number of times it is counted is

$$k - \binom{k}{2} + \binom{k}{3} = \begin{cases} 1 - 0 + 0 = 1 & (k = 1), \\ 2 - 1 + 0 = 1 & (k = 2), \\ 3 - 3 + 1 = 1 & (k = 3). \end{cases}$$

Every region of $A \cup B \cup C$ is counted exactly once, and regions outside the union are not counted at all. So the right-hand side equals $\mathrm{P}(A \cup B \cup C)$.

(Equivalently, apply the two-event rule $\mathrm{P}(X \cup Y) = \mathrm{P}(X) + \mathrm{P}(Y) - \mathrm{P}(X \cap Y)$ to $X = A \cup B$ and $Y = C$, then to $(A \cap C) \cup (B \cap C)$.)

Each pudding gets at least one sixpence

Model. Stirring thoroughly and dividing into equal portions means that each sixpence lands in pudding 1, 2 or 3 with probability $\frac{1}{3}$ each, independently of the other sixpences. Take $r \geq 1$.

Let E_i be the event that pudding i contains no sixpence.

- E_1 happens when every one of the r sixpences avoids pudding 1. Each does so with probability $\frac{2}{3}$, so by independence $\mathrm{P}(E_1) = \left(\frac{2}{3}\right)^r$, and likewise for E_2, E_3.
- $E_1 \cap E_2$ happens when every sixpence goes into pudding 3, so $\mathrm{P}(E_1 \cap E_2) = \left(\frac{1}{3}\right)^r$, and likewise for the other pairs.
- $E_1 \cap E_2 \cap E_3$ would need all three puddings empty, which is impossible since $r \geq 1$. So its probability is 0.

By inclusion–exclusion, the probability that at least one pudding is empty is

$$p_r = \mathrm{P}(E_1 \cup E_2 \cup E_3) = 3\left(\tfrac{2}{3}\right)^r - 3\left(\tfrac{1}{3}\right)^r.$$

The probability that each pudding contains at least one sixpence is the complement:

$$1 - 3\left(\tfrac{2}{3}\right)^r + 3\left(\tfrac{1}{3}\right)^r = \frac{3^r - 3 \cdot 2^r + 3}{3^r}.$$

At least six sixpences

We need the values of r for which $p_r < \frac{1}{3}$. Direct calculation gives

$$p_5 = \frac{3 \cdot 32 - 3}{243} = \frac{93}{243} = \frac{31}{81} > \frac{27}{81} = \frac{1}{3}, \qquad p_6 = \frac{3 \cdot 64 - 3}{729} = \frac{189}{729} = \frac{7}{27} < \frac{9}{27} = \frac{1}{3}.$$

To be sure that no smaller r works, we show that p_r decreases. For $r \geq 1$,

$$p_{r+1} - p_r = 3\left(\tfrac{2}{3}\right)^r\left(\tfrac{2}{3} - 1\right) - 3\left(\tfrac{1}{3}\right)^r\left(\tfrac{1}{3} - 1\right) = -\left(\tfrac{2}{3}\right)^r + 2\left(\tfrac{1}{3}\right)^r = \left(\tfrac{1}{3}\right)^r(2 - 2^r),$$

which is 0 for $r = 1$ and negative for $r \geq 2$. So $p_1 = p_2 \geq p_3 \geq p_4 \geq p_5 = \frac{31}{81}$, and every $r \leq 5$ has $p_r > \frac{1}{3}$. (With $r = 0$ every pudding is certainly empty.) Hence the chance that some pudding has no sixpence is below $\frac{1}{3}$ only if $r \geq 6$; and since p_r keeps decreasing, every $r \geq 6$ does give $p_r \leq p_6 < \frac{1}{3}$. So the cook must use 6 or more sixpences, as required.

Two sixpences in each pudding

Now $r = 6$. Let T be the event that each pudding contains exactly two sixpences, and S the event that each pudding contains at least one. From the previous part,

$$\mathrm{P}(S) = 1 - p_6 = 1 - \frac{7}{27} = \frac{20}{27}.$$

There are $3^6 = 729$ equally likely ways to send the six (distinguishable) sixpences to the puddings. The number giving two to each pudding is the number of ways to split six objects into three labelled pairs, $\dfrac{6!}{2!\,2!\,2!} = 90$. So $\mathrm{P}(T) = \frac{90}{729} = \frac{10}{81}$.

Ask yourself Why is $\mathrm{P}(T \cap S)$ just $\mathrm{P}(T)$?

If every pudding has two sixpences then certainly every pudding has at least one, so $T \subseteq S$ and $T \cap S = T$. Therefore

$$\mathrm{P}(T \mid S) = \frac{\mathrm{P}(T)}{\mathrm{P}(S)} = \frac{10/81}{20/27} = \frac{10}{81} \cdot \frac{27}{20} = \frac{1}{6}.$$

Checks

For $r = 1$ and $r = 2$ the formula gives a probability 0 that every pudding has a sixpence, which is right: you can't fill three puddings with fewer than three coins. For $r = 3$ it gives $\frac{27-24+3}{27} = \frac{2}{9}$, which matches the direct count $3!/3^3 = 6/27$. An exact enumeration of all 3^r assignments for $r \leq 9$ agrees with the formula, confirms $p_5 = \frac{31}{81} \approx 0.383$ and $p_6 = \frac{7}{27} \approx 0.259$, and gives the final answer $\frac{1}{6} \approx 0.167$ (a simulation of 200 000 puddings gave 0.166).

Common slips

- Treating the sixpences as indistinguishable and counting the 28 ways of writing $6 = a + b + c$ as equally likely. They aren't: $(2, 2, 2)$ can happen in 90 ways, while $(6, 0, 0)$ can happen in only one.
- Checking only $r = 6$ and $r = 5$. "Must use 6 or more" also needs every smaller r to fail, which is why the monotonicity step matters.
- Forgetting that the triple intersection is empty, or giving it probability $(0)^r$ without comment.

Beyond STEP

The general inclusion–exclusion principle for n events, with alternating sums of all k-fold intersections, is proved by exactly the counting argument above, using $\sum_{j\geq 1}(-1)^{j-1}\binom{k}{j} = 1$. Applied to n puddings it counts *surjections*, and connects to Stirling numbers of the second kind and to the classic coupon-collector problem. Grimmett and Welsh, *Probability: An Introduction*, develops inclusion–exclusion and these occupancy problems.

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